PRAGMI Architecture: Complete Mathematical Reference

Every equation for every arrow, feedback loop, timing, and component

Amellia Mendel · Lisa Adler

April 17, 2026

Genesis Labs Research

Contents

1. Thalamic Gate (Input → Primary Sensory Cortex)
2. Dorsal/Ventral Stream Split (after Primary Sensory Cortex)
3. PFC + ACC → Thalamus Top-Down Feedback Loop
4. Hippocampus ↔ Association Cortex Memory Feedback Loop
5. Basal Ganglia Disinhibitory Gating Loop
6. Cerebellum Efference Copy + Error Loop
7. Neuromodulator Broadcasts (Global Operating-Mode Signals)
8. ACC Conflict Detection
9. Association Cortex Bidirectional Feedback
10. Primary Sensory Cortex (V1/A1)

A. Module-to-Biology Mapping

B. Log-Dynamic Timing Hierarchy

C. Append-Only Additions from Multi-Scale Temporal Homeostasis (MSTH)

D. Append-Only Additions from MorphSNN

E. Append-Only Additions from Early Childhood Brain Development Review

F. Append-Only Additions from Early-Life Sleep and Brain Development Review

1. Thalamic Gate (Input → Primary Sensory Cortex)

Arrow: Input → Thalamus → Primary Sensory Cortex (active gate, not passive relay)

$$y = x \odot \sigma(W_g, c + b_g) \tag{1}$$

$$g = \sigma(\beta_{\text{NE}} \cdot (W_g, c + b_g)) \tag{2}$$

Variables: x = raw sensory input; c = control signal from PFC/ACC (goals + conflict); β_{NE} = norepinephrine gain scalar; W_g, b_g = learnable gate parameters.

Timing: Top-down control signal c from PFC/ACC arrives before the sensory signal reaches V1/A1. NE scales precision in real time every cycle.

Halassa, M.M. and Kastner, S. (2017). "Thalamic functions in distributed cognitive control." *Nature Neuroscience*, 20(12), 1669–1679.
DOI: 10.1038/s41593-017-0020-1

2. Dorsal/Ventral Stream Split (after Primary Sensory Cortex)

Arrow: Primary Sensory Cortex → Dorsal Stream & Ventral Stream (parallel split)

$$\text{Ventral} = f_v(S) \quad \textit{(object identity, meaning, value)} \tag{3}$$

$$\text{Dorsal} = f_d(S) \quad \textit{(spatial location, motion, action guidance)} \tag{4}$$

Timing: Early split immediately after V1/A1. This split is a prerequisite for correct PFC subdivision. Without it, ventromedial PFC and dorsolateral PFC receive mismatched inputs and cannot specialize.

> Ungerleider, L.G. and Mishkin, M. (1982). "Two cortical visual systems." In Ingle, Goodale and Mansfield (Eds.), *Analysis of Visual Behavior*. MIT Press, 549–586.

3. PFC + ACC → Thalamus Top-Down Feedback Loop

Arrow: PFC + ACC → Thalamus (Layer 6 corticothalamic feedback)

$$c = \text{PFC}_{\text{goal}} + \beta \cdot \text{ACC}_{\text{conflict}} \tag{5}$$

Variables: c = composite control signal; PFC_{goal} = working memory / goal vector; $\text{ACC}_{\text{conflict}}$ = entropy-based conflict scalar; β = coupling weight.

Timing: Layer 6 corticothalamic feedback arrives before the next sensory cycle. This is a continuous loop, not a one-shot signal.

> Halassa, M.M. and Kastner, S. (2017). "Thalamic functions in distributed cognitive control." *Nature Neuroscience*, 20(12), 1669–1679.
> DOI: 10.1038/s41593-017-0020-1

4. Hippocampus ↔ Association Cortex Memory Feedback Loop

Arrow: Hippocampus ↔ Association Cortex (bidirectional replay and consolidation)

$$H \leftarrow H + \eta_{\text{fast}} \cdot f(\text{input}) \quad \textit{(fast encoding)} \tag{6}$$

$$A \leftarrow A + \eta_{\text{slow}} \cdot \alpha_{\text{ACh}} \cdot \text{replay}(H) \quad \textit{(slow consolidation)} \tag{7}$$

Variables: H = hippocampal sparse representation; A = association cortex representation; η_{fast} = high learning rate (encoding); η_{slow} = small learning rate (consolidation); α_{ACh} = acetylcholine modulation scalar.

Timing: Fast update during active input. Slow replay during offline and consolidation phases. ACh is high during encoding (write mode) and lower during replay (read mode).

> McClelland, J.L., McNaughton, B.L. and O'Reilly, R.C. (1995). "Why there are complementary learning systems in the hippocampus and neocortex." *Psychological Review*, 102(3), 419–457.
> DOI: 10.1037/0033-295X.102.3.419

5. Basal Ganglia Disinhibitory Gating Loop

Arrow: Basal Ganglia → Thalamus (disinhibitory selection, not excitation)

$$I_i \leftarrow I_i - \alpha \cdot S_i \cdot (1 + d) \quad \textit{(direct pathway: chosen channel)} \tag{8}$$

$$I_j \leftarrow I_j + \beta \cdot S_j \cdot (1 - d) \quad \forall\, j \neq i \quad \textit{(indirect pathway: competitors)} \tag{9}$$

Variables: I = tonic inhibition from GPi/SNr on thalamus; S = striatal activation for different actions; d = dopamine reward prediction error scalar.

Timing: DA-modulated selection happens before thalamic relay to cortex. The hyperdirect pathway provides fast broad suppression before fine selection begins.

> Mink, J.W. (1996). "The basal ganglia: focused selection and inhibition of competing motor programs." *Progress in Neurobiology*, 50(4), 381–425.
> DOI: 10.1016/S0301-0082(96)00042-1

6. Cerebellum Efference Copy + Error Loop

Arrow: Cortex → Cerebellum (efference copy) → Cortex (error refinement)

$$\text{pred} = f_{\text{cerebellum}}(\text{cmd},\, \text{state}) \tag{10}$$

$$\text{error} = \text{actual} - \text{pred} \tag{11}$$

$$\text{cmd}_{\text{next}} \leftarrow \text{cmd} - \eta \cdot \text{error} \tag{12}$$

Variables: cmd = cortical command (efference copy); pred = predicted sensory consequence; error = mismatch used for calibration.

Timing: Efference copy is sent before command execution. NE modulates the gain on error computation, especially under arousal or uncertainty.

> Wolpert, D.M., Miall, R.C. and Kawato, M. (1998). "Internal models in the cerebellum." *Trends in Cognitive Sciences*, 2(9), 338–347.
> DOI: 10.1016/S0166-2236(98)01312-4

7. Neuromodulator Broadcasts (Global Operating-Mode Signals)

Arrows: DA, NE, ACh, 5-HT broadcast to all modules simultaneously.

$$\Delta w \leftarrow \Delta w \cdot (1 + d) \quad \textit{(Dopamine: plasticity and reinforcement)} \tag{13}$$

$$g \leftarrow g \cdot \text{ne} \quad \textit{(Norepinephrine: arousal and gating gain)} \tag{14}$$

$$\text{input}_{\text{eff}} = \text{ach} \cdot \text{bottom-up} + (1 - \text{ach}) \cdot \text{top-down} \quad \textit{(Acetylcholine: balance)} \tag{15}$$

$$\text{exploration_rate} \propto (1 + s) \quad \textit{(Serotonin: uncertainty tolerance)} \tag{16}$$

Timing: All neuromodulators broadcast in real time and update every cycle. Without explicit neuromodulation, the system has fixed hyperparameters and cannot shift between operating modes.

> Yu, A.J. and Dayan, P. (2005). "Uncertainty, neuromodulation, and attention." *Neuron*, 46(4), 681–692.
> DOI: 10.1016/j.neuron.2005.04.026

8. ACC Conflict Detection

Arrow: Competing PFC activations → ACC → control signal to Thalamus and NE system

$$\text{conflict} = \sum_{i \neq j} |r_i \cdot r_j| \quad \text{or} \quad H(r) = -\sum_i p_i \log p_i \tag{17}$$

$$\text{control_signal} = \beta \cdot \text{conflict} \tag{18}$$

Variables: r = vector of competing response activations; β = coupling coefficient to NE gain and thalamic gate precision.

Timing: Continuous scalar signal updated every cycle. Modulates thalamic gate gain and NE broadcast in real time. Functions as an uncertainty estimate, not a binary error flag.

Botvinick, M.M., Braver, T.S., Barch, D.M., Carter, C.S. and Cohen, J.D. (2001). "Conflict monitoring and cognitive control." *Psychological Review*, 108(3), 624–652.
DOI: 10.1037/0033-295X.108.3.624

9. Association Cortex Bidirectional Feedback

Arrow: Lower areas → Association Cortex ↔ Higher areas (MoE binding with feedback)

$$A = \text{MoE}(\text{concat}(S_1, S_2, \ldots, S_n) + \text{top_down_feedback}) \tag{19}$$

Variables: $S_1 \ldots S_n$ = inputs from different sensory streams; top_down_feedback = projection from PFC and ACC back into association cortex.

Constraint: MoE experts mimic regional specialization. The critical requirement is bidirectional connectivity: convergent input from lower areas and divergent feedback back down. Without the feedback path, the association cortex cannot participate in re-entrant loops.

Mesulam, M.M. (1998). "From sensation to cognition." *Brain*, 121(6), 1013–1052.
DOI: 10.1093/brain/121.6.1013

10. Primary Sensory Cortex (V1/A1)

Arrow: Thalamus → Primary Sensory Cortex (first cortical processing stage)

$$R(x,y) = G_{\sigma_c}(x,y) - G_{\sigma_s}(x,y) \tag{20}$$

Variables: G_{σ_c} = center Gaussian (excitatory); G_{σ_s} = surround Gaussian (inhibitory). Neurons fire strongly to preferred features (oriented edges, spatial frequencies) and weakly to uniform stimulation. Organized into orientation columns.

Note: No semantics or context at this stage. Raw feature extraction only. This is the input to the dorsal/ventral stream split.

Hubel, D.H. and Wiesel, T.N. (1968). "Receptive fields and functional architecture of monkey striate cortex." *Journal of Physiology*, 195(1), 215–243.
DOI: 10.1113/jphysiol.1968.sp008455

Summary: Implementation Priority

1. Thalamic Gate (highest priority: everything else depends on it)

2. Hippocampus feedback loop (resolves catastrophic forgetting)
3. Basal Ganglia disinhibitory gating (replaces softmax with biological selection)
4. Cerebellum efference copy loop (completes predictive coding)
5. ACC conflict monitoring (connects uncertainty to attentional precision)
6. Neuromodulator broadcasts (enables real-time operating mode shifts)

A. Module-to-Biology Mapping

Module	Biological Analog	Functional Role
ThalamicRouter	Thalamus	Multiplicative gating layer; primary input filter.
CLS_Module	Hippocampus / Cortex	Fast-path encoding and slow-path consolidation replay.
StriatalGatingSelector	Basal Ganglia	Disinhibitory selector; releases winning action from tonic bias.
CerebellarCorrection	Cerebellum	Parallel forward model; efference copy vs. sensory actuals.
ACC_Monitor	Anterior Cingulate	Entropy-based conflict detection modulating β_{NE}.

B. Log-Dynamic Timing Hierarchy

Cross-Frequency Coupling Rule: Gamma bursts are gated by the phase of the Theta cycle. Feature binding only occurs when the network is in an excitable state prepared by the slower rhythms.

Regime	Frequency	Functional Role
Delta	1–4 Hz	Master Scheduler: global state transitions and consolidation phases.
Theta	4–10 Hz	Phase Reference: sequential trajectory encoding in MemoryCortex.
Beta	13–30 Hz	Status Quo: maintains PFC goal states and suppresses irrelevant updates.
Gamma	30–100 Hz	Local Binding: high-frequency bursts for cortical feature unification.

C. Append-Only Additions from Multi-Scale Temporal Homeostasis (MSTH)

This appendix adds a multi-timescale homeostatic interpretation layer that can be attached to the existing PRAGMI architecture without altering Sections 1 through 10. The MSTH source describes coordinated regulation across ultrafast, fast, medium, and slow timescales, with a cross-scale coordinator embodying metaplasticity principles. New material is appended as additive support for homeostatic stability, not as a replacement for the original neuromodulator or timing sections.

Ultra-fast regulation

$$\text{Emergency}_t = \bigl(\sum_{k \in \{\text{mag, rate, var, mean}\}} E_k \geq 2\bigr) \wedge (t - t_{\text{last}} > \tau_{\text{refract}}) \tag{C1}$$

$$a_t^{\text{reg}} = a_t \odot (M_{\text{high}} \cdot 0.95 + M_{\text{low}}) \tag{C2}$$

Fast regulation

$$\varepsilon_{\text{Ca}} = \text{mean}(|c_t - c_{\text{target}}|) \tag{C3}$$

$$c_t^{\text{reg}} = c_t - 0.12 \cdot \sigma\bigl(4.0\,(c_t - c_{\text{target}})\bigr) \odot (c_t - c_{\text{target}}) \tag{C4}$$

Medium and slow regulation

$$A_{\text{accum}}(t) = A_{\text{accum}}(t-1) + A_{\text{level}}(t) \tag{C5}$$

$$\alpha_{\text{scale}} = \begin{cases} 0.996 & \text{if recent activity too high} \\ 1.004 & \text{if too low} \end{cases} \tag{C6}$$

$$P_{\text{recent}} = \tfrac{1}{3}\sum_{j=0}^{2} P_{\text{accumulator}}[-j-1] \tag{C7}$$

$$W_t^{\text{reg}} = 0.999 \times W_t \tag{C8}$$

Cross-scale coordination

$$H_{\text{system}} = \tfrac{1}{3}(H_{\text{activity}} + H_{\text{calcium}} + H_{\text{weights}}) \tag{C11}$$

$$\alpha_{\text{adaptive}} = 0.001 \times H_{\text{current}} \times H_{\text{stability}} \tag{C15}$$

$$W_{\text{final}}(t) = \text{slow_regulation}\bigl(\text{medium_regulation}(W_0(t),\, \rho(t)),\, P(t)\bigr) \tag{C17}$$

$$R_{\text{bio}} = 1.0 - \sum_{i=1}^{4} w_i \cdot |r_i^{\text{actual}} - r_i^{\text{expected}}| + B_{\text{bio}}^{\text{coord}} \tag{C18}$$

$$\mathbf{w} = [3.0,\; 0.3,\; 0.4,\; 0.4]^T \tag{C19}$$

> ***Interpretive note:*** *These relations naturally extend Section 7 and Appendix B by specifying an explicit hierarchy of regulatory timescales that may sit above real-time neuromodulator broadcasts and below long-horizon structural updates.*

> Multi-Scale Temporal Homeostasis (MSTH)
> `arXiv: 2602.07009v1`

D. Append-Only Additions from MorphSNN

This appendix adds a dynamic topology and diffusion layer relevant to PRAGMI's recurrent, multi-loop, brain-inspired design. MorphSNN introduces graph diffusion and spatio-temporal structural plasticity as append-only mathematical resources for dynamic routing, non-hierarchical lateral propagation, and instance-specific reorganization. Appended as optional support for association cortex and timing refinements.

Base spiking dynamics

$$u_i^{(t)} = \tau_{\text{decay}} \cdot u_i^{(t-1)} + C_i^{(t)} - V_{\text{th}} \cdot s_i^{(t-1)} \tag{D1}$$

$$s_i^{(t)} = \Theta\bigl(u_i^{(t)} - V_{\text{th}}\bigr) \in \{0, 1\} \tag{D2}$$

Graph diffusion and adaptive topology

$$H' = \sigma\bigl(\bar{D}^{-1/2}\, \hat{A}\, \bar{D}^{-1/2}\, H\, W\bigr) \tag{D3}$$

$$\hat{A}^{(t)} = \text{STSP}\bigl(S^{(t-1)},\, \{O_0^{(t)}, O_1^{(t-1)}, \ldots, O_{N-1}^{(t-1)}\}\bigr) \tag{D6}$$

$$\{I_1^{(t)}, \ldots, I_{N-1}^{(t)}\} = \text{GD}\bigl(O_0^{(t)},\, \hat{A}^{(t)}\bigr) \tag{D7}$$

Temporal traces and adjacency evolution

$$T_i^{(t)} = \lambda\, T_i^{(t-1)} + (1-\lambda)\, h_i^{(t)} \tag{D12}$$

$$S^{(t)} = \beta\, S^{(t-1)} + (1-\beta)\, \hat{A}^{(t)} \tag{D17}$$

$$\tilde{A}_{ij}^{(t)} = \begin{cases} S_{ij}^{(t)} & \text{if } S_{ij}^{(t)} \in \text{Top}_k(S_{i,:}^{(t)}) \\ 0 & \text{otherwise} \end{cases} \tag{D18}$$

Diffusion operator

$$P^{(t)} = D^{-1/2}\, \frac{\tilde{A}^{(t)} + (\tilde{A}^{(t)})^T}{2}\, D^{-1/2} \tag{D19}$$

$$I_i^{(t)} = \bigl[(P^{(t)})^M\, X^{(t)}\bigr]_i, \quad 1 \leq i < N \tag{D21}$$

Energy / smoothness

$$E_{\text{Dir}}(Y) = \text{tr}\bigl(Y^T\, L_{\text{norm}}\, Y\bigr) \tag{D22}$$

Interpretive note: *The MorphSNN graph diffusion links to biological nonsynaptic diffusion and STSP to structural plasticity. As an additive PRAGMI extension, this enriches Section 9 (Association Cortex Bidirectional Feedback) and Appendix B (Log-Dynamic Timing Hierarchy), especially where re-entrant signal flow and dynamic topology are desired.*

MorphSNN: Adaptive Graph Diffusion and Structural Plasticity for Spiking Neural Networks
`arXiv: 2603.14285v1`

E. Append-Only Additions from Early Childhood Brain Development Review

This appendix adds developmental support material relevant to why hierarchical specialization, sensory-first maturation, and later association-cortex refinement are biologically coherent. The supplied review introduces no numbered equations; this appendix is prose-only interpretive support.

Brain volume at birth is approximately 35% of adult size, doubles in the first year, and reaches roughly 80% by age two. Cortical grey matter volume increases 108–149% in the first year and 14–19% in the second. Surface area expands 76% in the first year and 22% in the second. Resting-state functional networks for sensorimotor, visual, and auditory systems are present at term. The default-mode network core structure is present by approximately six months. Language network asymmetry emerges by around two years. Myelination proceeds sensory before motor, projection before associative, and posterior before anterior.

Interpretive note: *These developmental observations support the original PRAGMI separation of primary sensory processing, stream specialization, and later integration. They also reinforce a staged maturation view in which early structural and functional scaffolds precede later reorganization and fine-tuning.*

Gilmore, J.H., Knickmeyer, R.C. and Gao, W. "Imaging structural and functional brain development in early childhood." *Nature Reviews Neuroscience* (nihms970191).

F. Append-Only Additions from Early-Life Sleep and Brain Development Review

This appendix adds sleep-dependent developmental material that aligns with the original hippocampus–association cortex replay and consolidation loop (Section 4, Appendix B). The supplied review contains no numbered equations; this appendix preserves it as prose support for the existing memory-consolidation and timing hierarchy sections.

Sleep plays a critical role in brain development. Topographical shifts in maximal slow wave activity from posterior to anterior regions reflect cortical maturation. Sleep spindles and slow wave activity are linked to synaptic plasticity, myelination, and cognitive development. Napping may be a marker of hippocampal maturation: children who nap may have less developed hippocampi and therefore require frequent offline consolidation. The transition out of napping may be associated with changes in sleep need, neurocognitive function, and memory system maturation.

> ***Interpretive note:*** *This material strengthens Section 4 and Appendix B by providing a developmental sleep basis for replay, consolidation, and oscillatory timing. Slow wave sleep, spindles, REM, and nap dependence are all directly tied to maturation, myelination, cortical plasticity, and memory consolidation.*

> Lokhandwala, S. and Spencer, R.M.C. "Relations between sleep patterns early in life and brain development: A review."